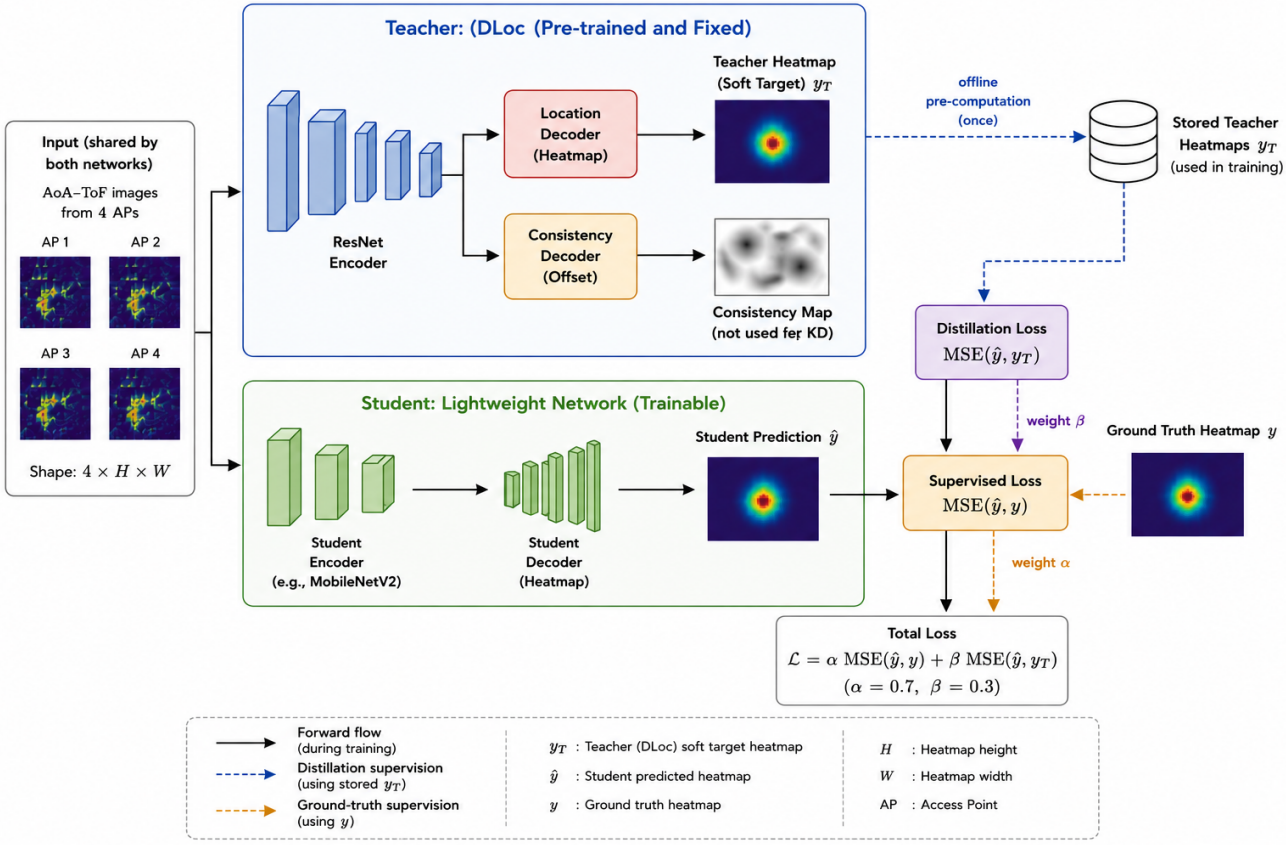

Teacher: (DLoc (Pre-trained and Fixed)
Input (shared by both networks)
AoA–ToF images from 4 APs
AP 1
AP 2
AP 3
AP 4
Shape: $4 \times H \times W$
ResNet Encoder
Location Decoder (Heatmap)
Consistency Decoder (Offset)
Teacher Heatmap (Soft Target) y_T
Consistency Map (not used for KD)
offline pre-computation (once)
Stored Teacher Heatmaps y_T (used in training)
Student: Lightweight Network (Trainable)
Student Encoder (e.g., MobileNetV2)
Student Decoder (Heatmap)
Student Prediction $\hat{y}$
Distillation Loss $\mathrm{MSE}(\hat{y}, y_T)$
weight β
Ground Truth Heatmap y
Supervised Loss $\mathrm{MSE}(\hat{y}, y)$
weight α
Total Loss
$\mathcal{L} = \alpha\ \mathrm{MSE}(\hat{y}, y) + \beta\ \mathrm{MSE}(\hat{y}, y_T)$
$(\alpha = 0.7,\ \beta = 0.3)$
Forward flow (during training)
Distillation supervision (using stored y_T)
Ground-truth supervision (using y)
y_T : Teacher (DLoc) soft target heatmap
$\hat{y}$: Student predicted heatmap
y : Ground truth heatmap
H : Heatmap height
W : Heatmap width
AP : Access Point